

Homework 03

Question 1: Build an `explain-to-me` skill

Create a small skill named `explain-to-me` that helps you understand a homework question.

1. Design your `SKILL.md` file so that, when you explicitly ask your agent to use the skill to read a homework question, the agent communicates with you and explains what the question is asking.
2. Keep the skill file short. It should contain a few useful rules, instructions, and a clear statement of your intent. For example, you may ask the agent to explain the goal of the question, clarify what work is required, and point you to an appropriate section of the Week 3 lectures: [Ridge Regression: Stability Through Shrinkage](#) or [From a Penalized Objective to a Fitted Ridge Model](#). The skill should help you understand the question without completing the solution for you.
3. Create the skill yourself or with the help of AI. Save it as `explain-to-me/SKILL.md`. Then ask your AI agent to install the skill using its normal skill-installation method. Test it by explicitly asking the agent to use `explain-to-me` to read and explain Question 2.
4. If the skill does not work well, or if it begins to disturb your usual workflow, ask your agent to remove the installed skill. You may revise and reinstall it if you wish.
5. In all cases, append the complete contents of your developed `explain-to-me/SKILL.md` file at the end of your Homework 3 submission under the heading `explain-to-me skill`.

ANSWER: Bottom of page

Question 2: Ridge regression with nearly collinear predictors

Let $n = 100$. Generate mutually independent standard normal random variables

$$L_i, U_{i1}, U_{i2}, X_{i3}, X_{i4}, \quad i = 1, \dots, n,$$

and define

$$X_{i1} = L_i + 0.04U_{i1}, \quad X_{i2} = L_i + 0.04U_{i2}.$$

Center and standardize each column of $\mathbf{X}$ to have mean zero and average squared value one, following the [lecture's scaling convention](#).

Continue to use $\mathbf{X} \in \mathbb{R}^{n \times p}$ for the standardized matrix and keep it fixed throughout the question, where $p = 4$ is the number of predictors. The unpenalized intercept gives $p + 1 = 5$ total fitted coefficients. Let

$$\boldsymbol{\beta} = (1.5, 1.5, 1, 0)^\top, \quad \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}.$$

Generate a training response and an independent test response at the same predictor values:

$$\begin{aligned} \mathbf{y} &= \boldsymbol{\mu} + \boldsymbol{\epsilon}, \\ \mathbf{y}^* &= \boldsymbol{\mu} + \boldsymbol{\epsilon}^*, \end{aligned}$$

where

$$\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^* \stackrel{\text{ind}}{\sim} \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n).$$

Center the training response as $\bar{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}_n$. For each $\lambda \in \{0, 0.02, 0.2\}$, fit the slopes using the lecture's ridge formula:

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \bar{\mathbf{y}}.$$

The unpenalized intercept is $\hat{\beta}_0 = \bar{y}$; add this training mean back when predicting either response. The case $\lambda = 0$ is ordinary least squares. Use seed `43231` before generating the design and responses. R and Python use different random-number generators, so their exact numerical results need not agree.

- a. Generate $\mathbf{X}$ once. Report the correlation between its first two columns and its largest and smallest singular values. Use these quantities to explain why the estimator contrast $\hat{\beta}_1 - \hat{\beta}_2$ should be much more variable than $\hat{\beta}_1 + \hat{\beta}_2$.
- b. Independently repeat the generation of $\mathbf{y}$ and $\mathbf{y}^*$, followed by all three fits, 200 times while keeping $\mathbf{X}$ fixed. Draw the training and test noise inside each repetition, then reuse that realized pair for all three values of λ . For each value of λ , report the empirical mean and standard deviation of

$$\hat{\beta}_1, \quad \hat{\beta}_2, \quad \hat{\beta}_1 + \hat{\beta}_2, \quad \hat{\beta}_1 - \hat{\beta}_2.$$

Also report the average training MSE and the average test MSE calculated from the independent test response, where

$$\text{MSE}_{\text{test}} = \frac{1}{n} \|\mathbf{y}^* - \hat{\beta}_0 \mathbf{1}_n - \mathbf{X} \hat{\boldsymbol{\beta}}\|_2^2.$$

- c. Interpret the coefficient estimates and test MSE using bias and variance. Explain why ridge can substantially stabilize the individual coefficients without producing an equally large change in the fitted values.

ANSWER: The first two predictors are almost identical. Their sum is a well-observed direction because both columns change together. Their difference is a weak direction because $\mathbf{x}_1 - \mathbf{x}_2$ is nearly zero. A small singular value records this lack of information.

The following code generates the design once and changes only the two response vectors across the 200 repetitions.

► Show the workflow

correlation largest_singular_value smallest_singular_value

0.9977 14.1415 0.4754

► Show the workflow

Empirical mean (standard deviation) across 200 repetitions

lambda	beta1	beta2	sum	difference
0.00	1.496 (1.544)	1.510 (1.549)	3.005 (0.095)	-0.014 (3.091)
0.02	1.487 (0.162)	1.489 (0.166)	2.976 (0.094)	-0.002 (0.314)
0.20	1.368 (0.046)	1.369 (0.047)	2.737 (0.086)	-0.001 (0.034)

► Show the workflow

Average training and test MSE

lambda	training_mse	test_mse
0.00	0.965	1.062
0.02	0.975	1.049
0.20	1.077	1.133

The correlation is about 0.998. The smallest singular value is approximately 0.4 to 0.5, whereas the largest is about 14. The design therefore contains much less information in its weakest direction.

The individual OLS coefficients have large standard deviations, and the coefficient difference is particularly unstable. In contrast, the estimated sum has a standard deviation of about 0.1. Ridge with $\lambda = 0.02$ sharply reduces the variability of the individual coefficients and the difference while changing the sum relatively little.

The stronger penalty $\lambda = 0.2$ reduces variance further, but it also shrinks the coefficient sum noticeably below its true value of 3. That additional bias raises the test MSE in both language implementations. The intermediate penalty gives a slightly smaller average test MSE than OLS in this simulation.

Question 3: What does ridge regression shrink?

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered predictor matrix, and let $\mathbf{\bar{y}} \in \mathbb{R}^n$ be the centered response. Use the supplied predictor scales for this algebraic question; the columns need not have unit variance. Here p counts predictors, $\boldsymbol{\beta} \in \mathbb{R}^p$ contains their slopes, and $p + 1$ counts all fitted coefficients, including the separately fitted, unpenalized intercept. Ridge regression minimizes

$$L_{\lambda}(\boldsymbol{\beta}) = \frac{1}{2n} \| \mathbf{\bar{y}} - \mathbf{X}\boldsymbol{\beta} \|^2 + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2.$$

a. Derive the ridge normal equation

$$(\mathbf{X}^T \mathbf{X} + n\lambda \mathbf{I}_p) \hat{\boldsymbol{\beta}}_{\lambda} = \mathbf{X}^T \mathbf{\bar{y}}.$$

Explain why it has a unique solution whenever $\lambda > 0$, even if $\mathbf{X}$ is rank deficient or $p > n$.

b. Let

$$\mathbf{X} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^T$$

be a compact singular value decomposition, where $r = \text{rank}(\mathbf{X})$, $\mathbf{U}_r \in \mathbb{R}^{n \times r}$, $\mathbf{D}_r \in \mathbb{R}^{r \times r}$, and $\mathbf{V}_r \in \mathbb{R}^{p \times r}$. Show that ridge multiplies the fitted response component in the j th left singular-vector direction, for $j = 1, \dots, r$, by

$$\rho_j(\lambda) = \frac{d_j^2}{d_j^2 + n\lambda}.$$

Suppose $n = 100$, $\lambda = 0.01$, and the nonzero singular values are

$$d_1 = 10, \qquad d_2 = 2, \qquad d_3 = 0.2.$$

Calculate the three shrinkage factors and the total effective degrees of freedom, including the unpenalized intercept:

$$\text{df}_{\text{eff}}(\lambda) = 1 + \sum_{j=1}^3 \rho_j(\lambda).$$

c. Which direction receives the strongest shrinkage? Explain why shrinking this direction can reduce prediction variance. Under what circumstance could the same shrinkage produce substantial prediction bias?

ANSWER: Differentiate the objective with respect to $\boldsymbol{\beta}$:

$$\nabla L_{\lambda}(\boldsymbol{\beta}) = -\frac{1}{n} \mathbf{X}^T (\mathbf{\bar{y}} - \mathbf{X}\boldsymbol{\beta}) + \lambda \boldsymbol{\beta}.$$

Setting the gradient equal to zero gives

$$\frac{1}{n} \mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}}_{\lambda} + \lambda \hat{\boldsymbol{\beta}}_{\lambda} = \frac{1}{n} \mathbf{X}^T \mathbf{\bar{y}}.$$

Multiplying by $\mathbf{n}$ gives the stated normal equation and hence

$$\hat{\boldsymbol{\beta}}_{\lambda} = (\mathbf{X}^{\top} \mathbf{X} + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}^{\top} \bar{\mathbf{y}}.$$

For every nonzero vector $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^{\top} (\mathbf{X}^{\top} \mathbf{X} + n\lambda \mathbf{I}_p) \mathbf{a} = \|\mathbf{X}\mathbf{a}\|_2^2 + n\lambda \|\mathbf{a}\|_2^2 > 0$$

when $\lambda > 0$. The matrix is therefore positive definite and invertible, even when $\mathbf{X}^{\top} \mathbf{X}$ is singular.

Using the singular value decomposition,

$$\hat{\boldsymbol{\beta}}_{\lambda} = \mathbf{V}_r (\mathbf{D}_r^2 + n\lambda \mathbf{I}_r)^{-1} \mathbf{D}_r \mathbf{U}_r^{\top} \bar{\mathbf{y}}.$$

If $r < p$, directions orthogonal to the columns of $\mathbf{V}_r$ lie in the null space of $\mathbf{X}$. Ridge sets those coefficient components to zero because they increase the penalty without changing the fitted response.

Multiplying by $\mathbf{X} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^{\top}$ gives

$$\mathbf{X} \hat{\boldsymbol{\beta}}_{\lambda} = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + n\lambda} \mathbf{u}_j \mathbf{u}_j^{\top} \bar{\mathbf{y}}.$$

Thus, the fitted response component in direction $\mathbf{u}_j$ is multiplied by $\rho_j(\lambda)$. Here $n\lambda = 1$, so

$$\begin{aligned} \rho_1 &= \frac{100}{101} \approx 0.9901, \\ \rho_2 &= \frac{4}{5} = 0.8, \\ \rho_3 &= \frac{0.04}{1.04} \approx 0.0385. \end{aligned}$$

Including the intercept,

$$\text{df}_{\text{eff}}(0.01) = 1 + 0.9901 + 0.8 + 0.0385 \approx 2.8286.$$

The third direction receives the strongest shrinkage. Its small singular value means that the data provide little information in that direction, so an unregularized estimate can have high variance. Shrinkage reduces this sensitivity to response noise. However, if the true mean response has a large component in the third left singular-vector direction, reducing that component to about 3.85% of its least-squares value can introduce substantial prediction bias.

Question 4: Ridge regression and optimization

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered predictor matrix with $p = 2$. Its columns need not have unit variance in this question. The intercept is fitted separately and is not penalized. Suppose the two eigenvalues of

$$\mathbf{A}_0 = \frac{1}{n} \mathbf{X}^{\top} \mathbf{X}$$

are 9 and 0.01. Ridge regression with $\lambda = 0.09$ has curvature matrix

$$\mathbf{A}_{\lambda} = \frac{1}{n} \mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}_p.$$

a. Calculate the condition numbers of $\mathbf{A}_0$ and $\mathbf{A}_{\lambda}$. Explain what the change says about the geometry of the ridge objective.

b. Gradient descent uses

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} - \eta \nabla L_{\lambda}(\boldsymbol{\beta}^{(k)}).$$

Let $\hat{\boldsymbol{\beta}}_{\lambda}$ be the ridge minimizer and define the optimization error

$$\mathbf{e}^{(k)} = \boldsymbol{\beta}^{(k)} - \hat{\boldsymbol{\beta}}_{\lambda}.$$

Show that the component of $\mathbf{e}^{(k)}$ in an eigendirection with unpenalized eigenvalue a is multiplied at each step by

$$1 - \eta(a + \lambda).$$

Let $M = 9.09$, the largest eigenvalue of $\mathbf{A}_{\lambda}$. Find the interval of step sizes that contracts every eigendirection. Then compare these two choices:

- For $\eta = 1/M$, calculate the two contraction factors and explain which direction converges faster.
- For $\eta = 2.1/M$, determine whether the method converges. Assume the initial error has a nonzero component in the eigendirection corresponding to M .

c. Increasing λ generally improves the condition number. Explain why choosing λ to make gradient descent converge as quickly as possible is not a valid statistical tuning rule.

ANSWER: The condition number is the ratio of the largest to the smallest eigenvalue. Without ridge,

$$\kappa(\mathbf{A}_0) = \frac{9}{0.01} = 900.$$

Adding $\lambda = 0.09$ changes the two eigenvalues to

$$9 + 0.09 = 9.09 \quad \text{and} \quad 0.01 + 0.09 = 0.10.$$

Therefore,

$$\kappa(\mathbf{A}_{\lambda}) = \frac{9.09}{0.10} = 90.9.$$

The ridge objective is still more curved in one direction than the other, but the disparity is much smaller. Its contours are less elongated, which improves numerical conditioning.

Let $\hat{\beta}_\lambda$ denote the ridge minimizer and define

$$\mathbf{e}^{(k)} = \boldsymbol{\beta}^{(k)} - \hat{\beta}_\lambda.$$

Because the ridge objective is quadratic,

$$\nabla L_\lambda(\boldsymbol{\beta}^{(k)}) = \mathbf{A}_\lambda \mathbf{e}^{(k)}.$$

The error update is consequently

$$\mathbf{e}^{(k+1)} = (\mathbf{I}_p - \eta \mathbf{A}_\lambda) \mathbf{e}^{(k)}.$$

In an eigendirection whose unpenalized eigenvalue is a , $\mathbf{A}_\lambda$ has eigenvalue $a + \lambda$. The corresponding error component is therefore multiplied by $1 - \eta(a + \lambda)$.

Every direction contracts when

$$|1 - \eta(a + \lambda)| < 1$$

for every eigenvalue. It is enough to impose

$$0 < \eta < \frac{2}{M} = \frac{2}{9.09} \approx 0.2200.$$

For $\eta = 1/M$, the contraction factors are

$$1 - \frac{9.09}{9.09} = 0$$

in the strong direction and

$$1 - \frac{0.10}{9.09} \approx 0.9890$$

in the weak direction. The strong-direction error disappears in one step, but the weak-direction error decreases slowly.

For $\eta = 2.1/M$, the strong-direction factor is

$$1 - \frac{2.1(9.09)}{9.09} = -1.1.$$

Its magnitude exceeds one, so a nonzero initial error component in that eigendirection alternates sign and grows. Under the condition stated in the question, the method does not converge even though the weak-direction component may still contract.

The penalty defines the statistical estimator. A larger value can reduce variance and improve conditioning, but it also introduces more shrinkage bias. The step size and numerical algorithm should be chosen to solve the objective accurately. The penalty should be chosen using statistical information about prediction, such as cross-validation or GCV, rather than computational speed.

Question 5: Selecting the ridge penalty for real-estate prediction

The supplied [data/realstate.csv](#) contains 414 real-estate transactions from Sindian District, New Taipei City, Taiwan. The quantitative response [price](#) is the unit house price. The six predictors are the transaction date, house age, distance to the nearest mass rapid transit station, number of nearby convenience stores, latitude, and longitude. Treat all six predictors as quantitative in this question. The column [row_id](#) identifies an observation and must not be used as a predictor.

The file [data/realstate-split-folds.csv](#) assigns 332 observations to the training set and 82 observations to the final test set. It also gives ten fold labels for the training observations. Match the two files using [row_id](#) and do not use the test responses while selecting a model.

Use the penalty grid

$$\Lambda = \{0\} \cup \{10^{-4+0.05l} : l = 0, \dots, 120\}.$$

For every fit, center and standardize each predictor using only the observations available to that fit, with the variance divided by the number of fitting observations. Center the response and leave the intercept unpenalized.

- Perform ten-fold cross-validation over Λ using the supplied training folds. Estimate the predictor means and scales separately within each set of nine training folds. For each λ , calculate the mean validation MSE and $\text{SE}(\lambda)$, the sample standard deviation of the ten fold errors divided by $\sqrt{10}$. Use this SE for the conventional one-standard-error heuristic. Plot the mean validation MSE and identify $\lambda_{\min}$. Choose λ_{1se} as the largest candidate in Λ , including OLS, satisfying

$$\text{MSE}(\lambda) \leq \text{MSE}(\lambda_{\min}) + \text{SE}(\lambda_{\min}).$$

- Use all $n = 332$ training observations to calculate the [lecture's GCV criterion](#) for each $\lambda \in \Lambda$:

$$\text{GCV}(\lambda) = \frac{\text{MSE}_{\text{train}}(\lambda)}{\{1 - \text{df}_{\text{eff}}(\lambda)/n\}^2}.$$

Here $\text{MSE}_{\text{train}}$ is the training MSE, and df_{eff} includes the unpenalized intercept, as in Question 3. Estimate means and scales once from all training observations and keep them fixed for this calculation. Plot the GCV curve and identify λ_{GCV} .

- Fix all three choices before examining the test responses. Refit OLS and ridge at $\lambda_{\min}$, λ_{1se} , and λ_{GCV} using all training observations. Report each model's training MSE, test MSE, and effective degrees of freedom. Interpret the differences and explain why the model with the smallest observed test MSE must not be selected after viewing this table.

Starter workflow

After matching the two files by [row_id](#), use this order:

```
for each validation fold:
  estimate predictor means and scales from the other nine folds
  for each lambda:
    fit on those nine folds and record validation MSE
summarize the ten fold errors and fix lambda_min and lambda_1se
compute GCV using all training observations and fix lambda_GCV
only then refit the four reported models and evaluate the test responses
```

ANSWER: The ridge fitting function below performs three statistical operations. It estimates the predictor means and scales from the fitting observations, solves the ridge normal equation on the standardized scale, and transforms the coefficients back to the original predictor scale for prediction. It is used for the final refits. In the cross-validation loop, each fold's means, scales, and response center are computed once and reused across every penalty. In the GCV loop, the full training means, scales, and response center are likewise computed once and reused across the full penalty grid.

Write $\bar{x}_j$ and s_j for the fitting-sample mean and scale of raw predictor j , and $\bar{y}$ for the fitting-sample response mean. The code uses `x_bar`, `s`, and `y_bar` for these quantities, `X` for standardized predictors, and `beta_hat` for their fitted slopes. Conversion to raw-scale coefficients is

$$\hat{\beta}_{\text{raw},j} = \frac{\hat{\beta}_j}{s_j}, \quad \hat{\beta}_{\text{raw},0} = \bar{y} - \sum_{j=1}^p \bar{x}_j \hat{\beta}_{\text{raw},j}.$$

The code calls these `beta_hat_raw` and `raw_intercept`. Both representations give the same predictions.

For GCV, use the training MSE and the total effective degrees of freedom. With $n = 332$ training observations and $p = 6$ predictors, the singular values d_j of the standardized training matrix give

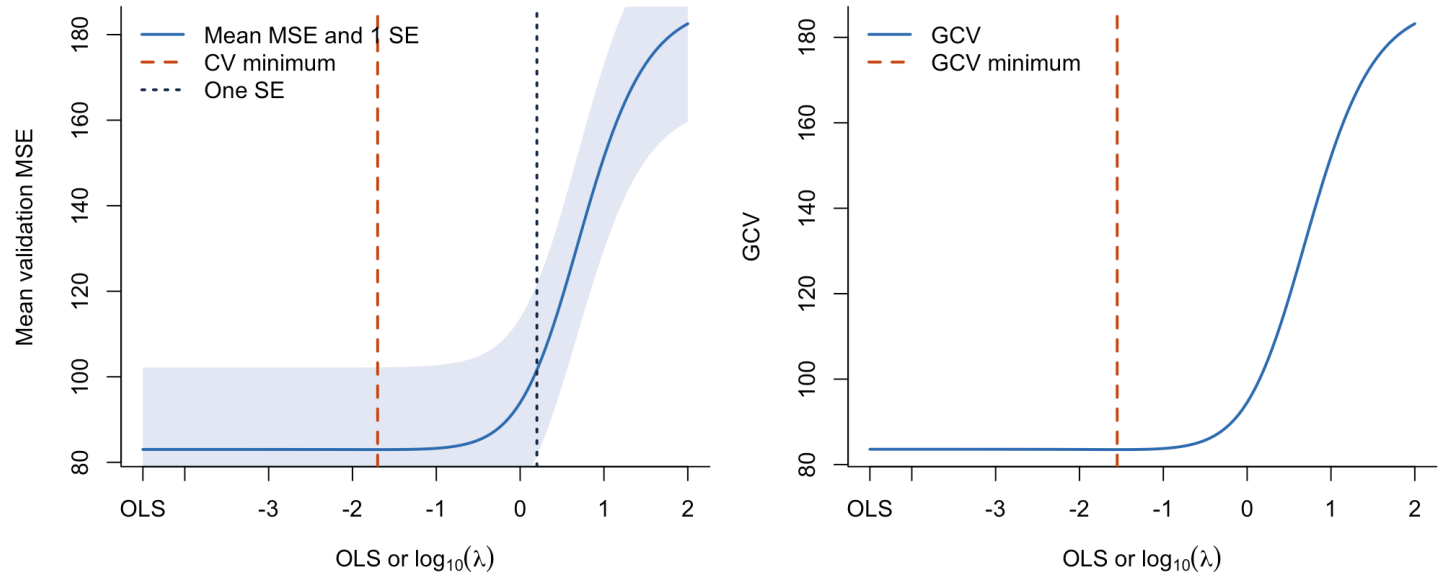
$$\text{df}_{\text{eff}}(\lambda) = 1 + \sum_{j=1}^p \frac{d_j^2}{d_j^2 + n\lambda}.$$

The leading one accounts for the unpenalized intercept.

The quantity called `cv_se` below is the conventional one-standard-error summary across folds. Because the ten fitted training sets overlap, it is a useful model-selection heuristic, not an independent-sample standard error.

► Show the workflow

► Show the workflow



Training-only penalty selection by ten-fold cross-validation and GCV.

► Show the workflow

► Show the workflow

```
lambda_min  lambda_1se  lambda_gcv one_se_limit
0.0199526   1.5848932   0.0281838  102.2075631
```

► Show the workflow

Final training and test comparison				
rule	lambda	df_eff	train_mse	test_mse
OLS	0.00000	7.000	80.094	66.322
CV minimum	0.01995	6.773	80.135	66.000
One SE	1.58489	3.033	100.374	81.200
GCV	0.02818	6.689	80.171	65.903

The three training-only rules select

$\lambda_{\min} = 0.01995, \quad \lambda_{1se} = 1.58489, \quad \lambda_{GCV} = 0.02818.$

The minimum-CV and GCV choices are close. They retain effective degrees of freedom of approximately 6.77 and 6.69, respectively, including the intercept. The one-standard-error rule chooses much stronger regularization and reduces the effective degrees of freedom to about 3.03. The fold errors vary substantially, so the one-standard-error threshold permits a wide range of penalties.

The final comparison is approximately

Rule	λ	Effective df	Training MSE	Test MSE
OLS	0	7.000	80.094	66.322
CV minimum	0.01995	6.773	80.135	66.000
One SE	1.58489	3.033	100.374	81.200
GCV	0.02818	6.689	80.171	65.903

OLS has the smallest training MSE, as expected. The minimum-CV and GCV ridge fits use mild shrinkage and have very similar test errors. The one-standard-error model has much stronger shrinkage and a larger test MSE on this particular test set.

This observed ordering does not invalidate the one-standard-error rule and does not authorize choosing GCV after seeing the test table. The three penalty rules were fixed using training information. The test set supplies one final assessment of those procedures. Selecting the smallest observed test MSE afterward would turn the test set into another validation set and make that minimum too favorable as an estimate of future performance.

GCV is also not an independent test result. It estimates prediction error from the training fit by correcting the training residuals using effective degrees of freedom. Finally, ridge controls coefficient variation within the specified linear model. It does not correct nonlinear structure, omitted variables, or other forms of model misspecification.

Reference

The real-estate data come from I-Cheng Yeh's [Real Estate Valuation dataset](#) in the UCI Machine Learning Repository. The distributed data retain the original observations and add only the course train-test and fold assignments.

explain-to-me skill

explain-to-me skill

```
---
name: explain-to-me
description: Explain a STAT 432 homework question when the student explicitly asks to use this skill, without solving it.
---

# Intent

Help the student understand what a homework question asks while leaving the mathematical and computational work to the student.

# Instructions

- Read the complete requested question before responding.
- Explain the goal, required work, expected outputs, and relevant notation in clear language.
- For Questions 2 and 3, point to a useful section of [Ridge Regression: Stability Through Shrinkage](https://teazrq.github.io/st
- Suggest a reasonable first step and ask what remains unclear.
- Do not perform calculations, derive results, write solution code, or provide the final answer.
```